

SPECIFICATION — NODE 3 — PAGE 4

EXTERNAL SESSION-BASED GEOMETRY SCANNER — NODE-SPECIFIC EXEMPLAR CLAIM-LIKE STATEMENTS (OPTIONAL, NON-LIMITING)

A. Overview

This page provides optional, non-limiting claim-like statements for Node 3 to support standalone protectability of the external session-based geometry scanner. A provisional application does not require claims. These statements are illustrative only and do not limit the disclosure.

B. Exemplary Node 3 System Statements (Non-Limiting)

1. A session-based scanning device comprising: a) a C-shaped housing or a substantially closed ring housing configured to be moved along a scan path during a scan session; b) at least one optical flow sensor configured to estimate incremental motion to derive a length estimate; c) at least one distance sensor selected from time-of-flight (ToF), LiDAR, and infrared (IR) distance sensors configured to derive a diameter estimate; and d) an inertial measurement unit (IMU) configured to support motion gating and artifact rejection; wherein a computing system selects valid scan segments using at least one quality index and outputs a privacy-preserving baseline geometry reference with a confidence indicator. 2. The device of statement 1, further comprising a pressure/contact sensor configured to confirm coupling during the scan session. 3. The device of statement 1, further comprising an optical sensor configured for photoplethysmography (PPG) to provide physiologic confirmation for anti-spoof scoring. 4. The device of statement 1, further comprising a temperature sensor configured to provide thermal context and/or compensate temperature-dependent drift in at least one sensing modality. 5. The device of statement 1, wherein the computing system computes a contact integrity index and a motion stability index and rejects scan segments not meeting threshold criteria. 6. The device of statement 1, wherein the computing system computes a distance plausibility index using cross-sensor agreement among multiple distance sensors. 7. The device of statement 1, wherein the computing system computes a repeatability index based on agreement among multiple scans and outputs a confidence indicator based on the repeatability index and at least one quality index. 8. The device of statement 1, wherein the baseline geometry reference includes zone-based profile summaries representing geometry variation along the scan path.

C. Exemplary Node 3 Method Statements (Non-Limiting)

9. A method comprising: a) acquiring optical flow sensor data during a guided scan session to estimate incremental motion; b) acquiring distance sensor data to estimate diameter; c) acquiring IMU data to gate motion and reject artifacts; d) computing at least one quality index to select valid scan segments; and e) outputting a privacy-preserving baseline geometry reference and a confidence indicator. 10. The method of statement 9, further comprising computing an anti-spoof confidence score using at least two modalities selected from contact sensing, IMU coherence, distance plausibility, thermal plausibility, and PPG-derived pulse features. 11. The method of statement 9, further comprising performing multiple scans and aggregating the valid scan segments to compute a consensus baseline reference. 12. The method of statement 9, further comprising prompting a user to repeat a scan when a quality index is below a threshold.

D. Optional Integration Statements (Non-Limiting)

13. The system or method of any prior statement, further comprising using the baseline geometry reference to normalize longitudinal time-series metrics generated by at least one other node under consent gating. 14. The system or method of any prior statement, further comprising time-aligning scan-derived baseline references with derived metrics from another node and computing an alignment score.

E. Variations

Any statement herein may be modified to include additional sensors, fewer sensors, substituted equivalent modalities, different housing geometries, different scan workflows, different duty cycles, and different processing distributions (on-node, on-device, cloud) while maintaining the functional intent of session-based baseline geometry characterization with validation, anti-spoof scoring, and confidence indicators.